Sheet 6

Parts and squeezed correlations

1. Let (e_j) and (f_k) be orthonormal bases of the two factors. For $\Psi = (e_0 \otimes f_0 + i e_1 \otimes f_1)/\sqrt{2}$ and $\rho = (2/3)\,|\Psi\rangle\langle\Psi| + (1/3)\,|e_2 \otimes f_0\rangle\langle e_2 \otimes f_0|$, compute both reduced states and their purities. Is the composite pure?
2. **Squeezed Gaussian (longer).** Let $s > 0$ and $r \geq 0$. Normalise

$$\psi_r \propto \exp\left[-\frac{e^{2r}(x_1 - x_2)^2}{4s^2} - \frac{e^{-2r}(x_1 + x_2)^2}{4s^2}\right].$$

   Use $u = (x_1 - x_2)/\sqrt{2}$, $v = (x_1 + x_2)/\sqrt{2}$ to compute the two collective variances and the local position and momentum variances. Check the local uncertainty product and evaluate the lecture's witness.

   > **First hint**
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   > Use the squared modulus for normalisation and position moments. The orthogonal change of coordinates has Jacobian of magnitude one, and it turns the density into a product of two one-dimensional Gaussians.

   > **Further hint**
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   > Write each local coordinate as a sum or difference of the new coordinates. Their cross moment vanishes. For momentum moments, apply the momentum derivatives to the wavefunction, rather than differentiating its squared modulus. Check the unsqueezed case before interpreting the collective result.

3. Independent classical shifts add variance n_x to each local position and n_p to each local momentum. Find the resulting collective witness. At $r = (\log 2)/2$, compare $n_x/s^2 = s^2 n_p/\hbar^2 = 0.1$ and 0.3. Does failure of this sufficient test prove separability?